HOSPITAL MANAGEMENT SYSTEM (HMS)

Requirements Traceability Matrix (RTM v2.0) — Part A & Part B

Bidirectional Implementation Traceability across Baseline Scope (STP v1.0) and Supplementary Sprint Scope

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1. Traceability Architecture & Structure

This Requirements Traceability Matrix (RTM) establishes full bidirectional traceability between the system requirements, development sprint activities, merged GitHub pull requests, source code implementations, and automated CTest verification targets.

To preserve complete academic and engineering integrity:

1. **Part A — Original STP Baseline RTM:** Retains the exact 54 requirements defined in the frozen baseline Software Test Plan (STP-HMS-2026-V1.0 Table 7). The denominator of 54 requirements (36 Functional, 10 Non-Functional, 8 Security) is strictly preserved.

2. **Part B — Additional Sprint Scope RTM:** Captures all supplementary functional extensions, non-functional quality improvements, and security hardening introduced during Sprint 1 and Sprint 2 via merged pull requests (#117, #118, #120, #121, #124, #127).

Execution Status Notice

Per Deliverable Part-3 guidelines, this document reflects test case design, requirement mapping, and implementation discovery. Execution status is formally tracked as **Not Executed** for the fresh Part-3 test campaign baseline, with historical execution evidence documented in the Remarks column.

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2. Part A — Original STP Baseline Requirements Traceability Matrix (54 Requirements)

The table below catalogs all 54 baseline requirements from STP v1.0 Table 7.

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-F-001	Register Patient with ID, name, age, gender, contact, address, medical note	TC-PAT-01	TC-PAT-01	TC-PAT-05_PatientDetailsAndRegistrationPersistence	tests/test_patient_details_and_list.cpp	Sprint 1	Issue #1 / PR #61, #80	Functional	System / Integration	Patient record created and immediately saved in patients.csv; valid confirmation displayed	**Covered**	Not Executed	Core registratio tested in test_pa nt_details_and_ .cpp under targe TC-PAT-05
S-F-002	Reject Duplicate Patient ID and leave stored records unchanged	TC-PAT-02	TC-PAT-02	TC-PAT-02_DuplicatePatientId	tests/test_duplicate_patient_id.cpp	Sprint 1	Issue #2 / PR #62	Functional	Unit / Integration	Error message "Patient ID already exists" displayed; duplicate rejected; CSV unchanged	**Covered**	Not Executed	Verified with case-insensitive IDs (e.g. p001 v P001)
S-F-003	Validate Patient Fields against rules VR-01 to VR-07 and VR-14	TC-PAT-03	TC-PAT-03	TC-PAT-03_ValidatePatientFields	tests/test_validate_patient_fields.cpp	Sprint 1 / 2	Issue #3, #119 / PR #63, #127	Functional / Boundary	Unit	Out-of-range inputs rejected with field-specific error; valid data accepted; immediate retry	**Covered**	Not Executed	Enhanced in Sprint 2 with immediate field validation and retry (PR #127)
S-F-004	Search Patients by Patient ID (exact) or name (case-insensitive partial)	TC-PAT-04	TC-PAT-04	TC-PAT-04_SearchPatients	tests/test_search_patients.cpp	Sprint 1	Issue #4 / PR #67, #70	Functional	Unit / System	Exact ID and partial name return matching records; whitespace trimmed; "No matching patient found" on miss	**Covered**	Not Executed	PR #70 added whitespace trimming verification
S-F-005	Display all stored fields of selected patient and list all registered patients	TC-PAT-05	TC-PAT-05	TC-PAT-05_PatientDetailsAndRegistrationPersistence	tests/test_patient_details_and_list.cpp	Sprint 1	Issue #5 / PR #68, #71	Functional	System	Displays all 7 stored fields for patient; list view shows all patients; "No patients registered" when empty	**Covered**	Not Executed	Shared test targ with immediate registration persistence
S-F-006	Update Patient Details (name, age, gender, contact, address, note) with validation	TC-PAT-06	TC-PAT-06	TC-PAT-06_UpdatePatientDetails	tests/test_update_patient_details.cpp	Sprint 1 / 2	Issue #27, #119 / PR #85, #127	Functional	System / Integration	Valid change reflected in next view and saved to patients.csv; invalid values rejected; same-field retry	**Covered**	Not Executed	Supports updati individual fields all fields with validation retry
S-F-007	Prevent Patient ID from being edited and reject update for nonexistent ID	TC-PAT-07	TC-PAT-07	TC-PAT-07_PreventPatientIdEdits	tests/test_prevent_patient_id_edits.cpp	Sprint 1	Issue #28 / PR #86	Functional	System	Update menu offers no Patient ID edit; updating nonexistent ID displays "Patient not found"	**Covered**	Not Executed	Enforces immutability of primary key BR-
S-F-008	Add Doctor with ID, name, specialization, department, contact, availability	TC-DOC-01	TC-DOC-01	TC-DOC-01_AddDoctor	tests/test_add_doctor.cpp	Sprint 1	Issue #6 / PR #64	Functional	System / Integration	Doctor record created and saved in doctors.csv; confirmation displayed	**Covered**	Not Executed	Enforces all 6 attributes of Doctor domain
S-F-009	Reject Duplicate Doctor ID and leave stored records unchanged	TC-DOC-02	TC-DOC-02	TC-DOC-02_RejectDuplicateDoctorId	tests/test_duplicate_doctor_id.cpp	Sprint 1	Issue #7 / PR #65	Functional	Unit / Integration	Error message "Doctor ID already exists" displayed; duplicate rejected; file unchanged	**Covered**	Not Executed	Enforces uniqueness rule BR-02

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-F-010	Validate Doctor Fields against rules VR-01, VR-02, VR-05, VR-08, VR-09, VR-14	TC-DOC-03	TC-DOC-03	TC-DOC-03_ValidateDoctorFields	tests/test_validate_doctor_fields.cpp	Sprint 1 / 2	Issue #8, #119 / PR #66, #127	Functional / Boundary	Unit	Invalid specialization, department, contact, or availability rejected; immediate retry on error	**Covered**	Not Executed	Enhanced in Sprint 2 with immediate field validation and retry (PR #127)
S-F-011	Search Doctors by Doctor ID (exact), name (partial), or specialization (partial)	TC-DOC-04	TC-DOC-04	TC-DOC-04_SearchDoctors	tests/test_search_doctors.cpp	Sprint 1	Issue #9 / PR #72	Functional	System	Exact ID and partial name/specialization return matches; "No matching doctor found" on miss	**Covered**	Not Executed	Case-insensitive matching across all 3 criteria
S-F-012	Display all stored fields of selected doctor and list all registered doctors	TC-DOC-05	TC-DOC-05	TC-DOC-05_ViewAndListDoctors	tests/test_view_list_doctors.cpp	Sprint 1	Issue #10 / PR #73	Functional	System	Displays complete doctor record (6 fields); list view shows all doctors; "No doctors registered" when empty	**Covered**	Not Executed	Validates complete doctor listing formatting
S-F-013	Update Doctor Details (name, specialization, department, contact, availability)	TC-DOC-06	TC-DOC-06	TC-DOC-06_UpdateDoctorDetails	tests/test_update_doctor_details.cpp	Sprint 1 / 2	Issue #29, #119 / PR #87, #127	Functional	System / Integration	Valid change reflected in next view and doctors.csv; invalid values rejected; same-field retry	**Covered**	Not Executed	Updated in Sprint 2 to support field-level validation and retry
S-F-014	Prevent Doctor ID from being edited and reject update for nonexistent ID	TC-DOC-07	TC-DOC-07	TC-DOC-07_PreventDoctorIdEdits	tests/test_prevent_doctor_id_edits.cpp	Sprint 1	Issue #30 / PR #88	Functional	System	Update menu offers no Doctor ID edit; updating nonexistent ID displays "Doctor not found"	**Covered**	Not Executed	Enforces immutability of primary key BR-
S-F-015	Book Appointment with system-generated ID (A0001+) and status Booked	TC-APT-01	TC-APT-01	TC-APT-01_BookAppointment	tests/test_book_appointment.cpp	Sprint 1	Issue #11 / PR #69	Functional	System / Integration	Valid booking displays Appointment ID, status Booked, and saves to appointments.csv	**Covered**	Not Executed	Validates sequential ID generation and initial status Booked
S-F-016	Reject Appointment Booking when Patient ID does not exist	TC-APT-02	TC-APT-02	TC-APT-02_RejectUnknownPatient	tests/test_reject_unknown_patient.cpp	Sprint 1	Issue #12 / PR #74	Functional	Integration	Booking with unregistered Patient ID displays "Patient not found"; no record created	**Covered**	Not Executed	Enforces foreign key integrity BR-03
S-F-017	Reject Appointment Booking when Doctor ID does not exist	TC-APT-03	TC-APT-03	TC-APT-03_RejectUnknownDoctor	tests/test_reject_unknown_doctor.cpp	Sprint 1	Issue #13 / PR #75	Functional	Integration	Booking with unregistered Doctor ID displays "Doctor not found"; no record created	**Covered**	Not Executed	Enforces foreign key integrity BR-04
S-F-018	Validate Appointment Date and Time against rules VR-10 and VR-11	TC-APT-04	TC-APT-04	TC-APT-04_ValidateAppointmentDateTime	tests/test_validate_appointment_date_time.cpp	Sprint 1 / 2	Issue #14, #119 / PR #76, #127	Functional / Boundary	Unit	Invalid date (2026-02-30) and time (24:00) rejected with error message; immediate retry	**Covered**	Not Executed	Verifies leap years, calendar boundaries, 24-hour clock

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-F-019	Reject Appointment Booking when Doctor has non-cancelled conflict (double-booking)	TC-APT-05	TC-APT-05	TC-APT-05_TC-NFR-08_PreventDoubleBooking	tests/test_prevent_double_booking.cpp	Sprint 2	Issue #23 / PR #91	Functional / Logic	Integration	Second booking for same doctor at same date/time displays "Doctor not available"; no record added	**Covered**	Not Executed	Enforces double-booking conflict rule BR-
S-F-020	Check Doctor Availability for given doctor, date, and time without modifying records	TC-APT-06	TC-APT-06	TC-APT-06_CheckDoctorAvailability	tests/test_check_doctor_availability.cpp	Sprint 2	Issue #24 / PR #92	Functional	System	Reports "Available" or "Not Available" accurately based on stored appointments; file unchanged	**Covered**	Not Executed	Standalone informational check (UC-10)
S-F-021	Exclude Cancelled appointments when determining doctor availability for booking	TC-APT-05	TC-APT-07	TC-APT-07_IgnoreCancelledAppointmentsInConflicts	tests/test_ignore_cancelled_appointments.cpp	Sprint 2	Issue #25 / PR #93	Functional / Logic	Integration	Re-booking doctor at same date and time succeeds after previous appointment was cancelled	**Covered**	Not Executed	Mapped to CTe target TC-APT- verifies BR-06 s reuse
S-F-022	View all appointments, or by Patient ID or Doctor ID, sorted by date then time	TC-APT-08	TC-APT-08	TC-APT-08_ViewAppointmentsDateTimeOrder	tests/test_view_appointments_date_time_order.cpp	Sprint 2	Issue #26 / PR #94	Functional	System	Appointments listed in ascending chronological order; displays "No appointments found" when empty	**Covered**	Not Executed	Note: Entity existence validation on filt is covered in supplementary TC-APT-12
S-F-023	Cancel Appointment by ID with staff confirmation; status changed to Cancelled	TC-APT-09	TC-APT-09	TC-APT-09_CancelAppointment	tests/test_cancel_appointment.cpp	Sprint 2	Issue #32, #120 / PR #95, #122	Functional	System / Integration	Status updated to Cancelled; record retained in CSV; confirmation workflow honored	**Covered**	Not Executed	Preserves audit record BR-06 / HMS-SR-006; summary promp added in PR #1
S-F-024	Reject Cancellation of nonexistent Appointment ID or already-cancelled appointment	TC-APT-10	TC-APT-10	TC-APT-10_RejectInvalidCancellation	tests/test_reject_invalid_cancellation.cpp	Sprint 2	Issue #33, #119 / PR #96, #127	Functional	System	Nonexistent ID displays "Appointment not found"; already-cancelled displays "already cancelled"	**Covered**	Not Executed	Retry loop adde in PR #127
S-F-025	Load records from patients.csv, doctors.csv, appointments.csv at startup	TC-PER-01	TC-PER-01	TC-PER-01_StartupCSVLoading	tests/test_startup_csv_loading.cpp	Sprint 1	Issue #15 / PR #77	Functional / Integration	System / Integration	With valid files containing N records each, system loads N records into memory on launch	**Covered**	Not Executed	Verifies startup parsing and rec instantiation BR-09
S-F-026	Treat missing CSV files as empty datasets and create file on next successful save	TC-PER-02	TC-PER-02	TC-PER-02_MissingCSVFiles	tests/test_missing_csv_files.cpp	Sprint 1	Issue #16 / PR #78	Functional / Robustness	System	Missing CSV treated as empty; application displays main menu without crash; file created on save	**Covered**	Not Executed	Satisfies gracef startup requirement BR-09 and HMS-NF-002
S-F-027	Skip malformed CSV rows during load, continue remaining lines, report skipped count	TC-PER-03	TC-PER-03	TC-PER-03_SkipMalformedCSV	tests/test_skip_malformed_csv.cpp	Sprint 1 / 2	Issue #17, #50 / PR #79, #112	Functional / Robustness	Integration	Skips corrupt lines and displays "X malformed record(s) skipped"; loads valid rows safely	**Covered**	Not Executed	PR #112 extend coverage to doc and appointmer CSV rows

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-F-028	Skip duplicate entity IDs and orphan appointments during startup loading	TC-PER-04	TC-PER-04	TC-PER-08_SkipDuplicateAndOrphanRecords	tests/test_skip_duplicate_and_orphan_records.cpp	Sprint 2	Issue #34 / PR #97	Functional / Integration	Integration	Duplicate IDs and orphan appointment rows ignored on load; included in skipped record count	**Covered**	Not Executed	Discrepancy: ST named TC-PER-04; registered in CTest as TC-PER-08
S-F-029	Write affected CSV file immediately after each successful modifying operation	TC-PER-05	TC-PER-05	TC-PAT-05_PatientDetailsAndRegistrationPersistence	tests/test_patient_details_and_list.cpp	Sprint 1	Issue #18 / PR #80	Functional / Integration	System / Integration	Immediately after each operation, corresponding CSV file contains the newly committed change	**Covered**	Not Executed	Discrepancy: ST named TC-PER-05; tested under tar TC-PAT-05
S-F-030	Make all records available after application is closed normally and restarted	TC-PER-06	TC-PER-06	TC-NFR-03_DataRetention	tests/test_data_retention.cpp	Sprint 1 / 2	Issue #19, #39 / PR #81, #101	Functional / Persistence	System	Data retained 100% across application close and restart; identical counts and field values	**Covered**	Not Executed	Discrepancy: ST named TC-PER-06; tested under tar TC-NFR-03
S-F-031	Report CSV write failure, remain running, and preserve previously valid state	TC-PER-07	TC-PER-07	TC-PER-07_HandleCSVWriteFailures	tests/test_csv_write_failures.cpp	Sprint 1 / 2	Issue #31, #51 / PR #89, #113	Functional / Reliability	Integration	Displays error message when write fails; previous file content remains valid; menu usable	**Covered**	Not Executed	PR #113 reinforced non-destructive atomic write guarantees (HMS-SR-005)
S-F-032	Display main menu and submenus with return option (Option 0)	TC-UI-01	TC-UI-01	TC-UI-01_MenuNavigation	tests/test_menu_navigation.cpp	Sprint 1	Issue #20 / PR #82	Functional	System	Main menu and 3 submenus navigate correctly; Option 0 returns to previous menu level	**Covered**	Not Executed	Verifies console menu hierarchy
S-F-033	Reject empty, non-numeric, or out-of-range menu selections with error message	TC-UI-02	TC-UI-02	TC-UI-02_RejectInvalidMenuOptions	tests/test_reject_invalid_menu_options.cpp	Sprint 1	Issue #21 / PR #83	Functional / Robustness	System	Inputs "9", "abc", "-1", and empty line produce error and re-display same menu without crashing	**Covered**	Not Executed	Satisfies input robustness for menu selection BR-08
S-F-034	Leave in-memory and stored records unchanged when any operation input fails	TC-UI-03	TC-UI-03	No dedicated CTest target; verified in tests/test_reject_invalid_menu_options.cpp and tests/test_csv_write_failures.cpp	tests/test_reject_invalid_menu_options.cpp, tests/test_csv_write_failures.cpp	Sprint 1	Issue #22 / PR #84	Functional / Integrity	System / Integration	Failed validation leaves affected CSV file byte-identical and in-memory records unchanged	**Covered**	Not Executed	Transactional state protection verified across menu option an file write failure test suites witho a dedicated standalone CTe executable targ
S-F-035	End every operation with either success confirmation or error message stating reason	TC-UI-04	TC-UI-04	TC-UI-04_OperationOutcomeMessages	tests/test_operation_outcome_messages.cpp	Sprint 2	Issue #35 / PR #98	Functional	System	Every operation in functional suite displays exactly one of the two clear outcomes	**Covered**	Not Executed	Verifies standardized us messaging

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-F-036	Terminate normally with exit code 0 when Option 4 Exit is selected	`TC-UI-05`	`TC-UI-05`	`TC-UI-05_ExitCleanly, TC-UI-05_ExitProcessCode`	`tests/test_exit_cleanly.cpp, tests/test_exit_process.cmake`	Sprint 2	Issue #36 / PR #99	Functional	System	Process terminates with exit code 0; all previously confirmed changes remain in data files	**Covered**	Not Executed	Two CTest targets: executable outp check and CMa exit code assert
S-NF-001	Input Crash Resilience: survive 20+ defined invalid keyboard input cases	`TC-NFR-01`	`TC-NFR-01`	`TC-NFR-01_InvalidInputCrashResilience`	`tests/test_invalid_input_crash_resilience.cpp`	Sprint 1	Issue #37 / PR #90	Reliability	System	0 crashes across 20 invalid input test cases; application displays error and returns to prompt	**Covered**	Not Executed	Exercises empty format strings, negative, extremely large numbers
S-NF-002	Robust Startup: start successfully for missing, empty, or corrupt file combinations	`TC-NFR-02`	`TC-NFR-02`	`TC-NFR-02_RobustStartup`	`tests/test_robust_startup.cpp`	Sprint 2	Issue #38 / PR #100	Reliability	System	Startup succeeds in all combinations (all missing, all empty, malformed rows); main menu displays	**Covered**	Not Executed	Verifies resilien against corrupte startup states
S-NF-003	Data Retention Rate: retain 100% of confirmed records after normal exit and restart	`TC-NFR-03`	`TC-NFR-03`	`TC-NFR-03_DataRetention`	`tests/test_data_retention.cpp`	Sprint 2	Issue #39 / PR #101	Persistence	System	Record counts and field values before exit equal those after restart (100% retention)	**Covered**	Not Executed	Verifies byte-lev integrity across application runs
S-NF-004	Load & Execution Latency: startup <= 3s, operations <= 2s with 1,000 synthetic records	`TC-NFR-04`	`TC-NFR-04`	`TC-NFR-04_Performance1000Records`	`tests/test_performance_1000_records.cpp`	Sprint 2	Issue #44 / PR #106, #125	Performance	Benchmark	Startup loading <= 3.0s; booking and search <= 2.0s with 1,000 records each	**Covered**	Not Executed	Registered with CTest via PR #125; measures in-memory / file I/O latency via d playStartupRec s()
S-NF-005	UI Usability & Menu Depth: prompts state format, errors state rule, menu depth <= 3	`TC-NFR-05`	`TC-NFR-05`	`TC-NFR-05_UsabilityMenuDepth`	`tests/test_usability_menu_depth.cpp`	Sprint 2	Issue #40 / PR #102	Usability	System	Checklist review passes 100%; menu depth of every operation <= 3 from main menu	**Covered**	Not Executed	Verifies navigational usability compliance
S-NF-006	Architectural Class Isolation: 5 OOP classes; file I/O restricted to FileManager	`TC-NFR-06`	`TC-NFR-06`	`TC-NFR-06_ArchitecturalClassIsolation`	`tests/test_architectural_class_isolation.cpp`	Sprint 2	Issue #42 / PR #104	Maintainability	Static / Architectural	Code review confirms 5 classes; file-stream usage found only in FileManager	**Covered**	Not Executed	Static source co inspection test target
S-NF-007	Cross-Platform Compilation: clean compilation with -std=c++17 on Windows and Linux	`TC-NFR-07`	`TC-NFR-07`	`No dedicated CTest executable target; verified via build configuration and CI`	`CMakeLists.txt, .github/workflows/linux-build.yml`	Sprint 2	Issue #45 / PR #107	Portability	Build	Clean compilation with -std=c++17 and clean build/test execution across Windows and Linux platforms	**Partial**	Not Executed	Discrepancy: Enforced at buil time via CMake compiler flags (-std=c++17) an CI workflow; lac a standalone runtime CTest t target executab

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-NF-008	Conflict Detection Accuracy: 100% accuracy for defined 4-scenario conflict matrix	TC-NFR-08	TC-NFR-08	TC-NFR-08_AppointmentConflictCorrectness	tests/test_appointment_conflict_correctness.cpp	Sprint 2	Issue #41 / PR #103	Correctness	Integration	All four matrix cases give the expected accept/reject result (100% accuracy)	**Covered**	Not Executed	Exercises same doctor/slot, diff time, diff doctor and post-cancellatio
S-NF-009	Standalone Deployment: run single-user offline without network sockets or external DB	TC-NFR-09	TC-NFR-09	TC-NFR-09_StandaloneOffline	tests/test_standalone_offline.cmake	Sprint 2	Issue #43, #54 / PR #105, #116	Deployment	System	Application runs with networking disabled; binary inspection proves zero network symbols	**Covered**	Not Executed	Verified via objdump symbo check in test_st dalone_offline.c ke
S-NF-010	CSV Interoperability: delimiter-separated files with headers, readable in text editor	TC-NFR-10	TC-NFR-10	TC-NFR-10_HumanReadableCSV	tests/test_human_readable_csv.cpp	Sprint 2	Issue #46 / PR #108	Interoperability	System	Each file opens in text editor and matches Table 3-3 layouts with exact header row	**Covered**	Not Executed	Validates comma-delimite columns and newline termination
S-SR-001	Accept-List Input Validation: validate keyboard inputs against Table 4-1 domains	TC-SEC-01	TC-SEC-01	TC-SEC-01_AcceptListInputValidation	tests/test_accept_list_input_validation.cpp	Sprint 2	Issue #47 / PR #109	Security	System	Boundary values accepted/rejected as defined; reject unauthorized characters	**Covered**	Not Executed	Enforces strict accept-list approach acros all fields
S-SR-002	CSV & Formula Injection Guard: reject commas, newlines, and prefixes (=, +, -, @)	TC-SEC-02	TC-SEC-02	TC-SEC-02_CSVFormulaInjectionGuard	tests/test_csv_formula_injection_guard.cpp	Sprint 2	Issue #48 / PR #110	Security	System	Delimiter and formula injection inputs rejected; no record stored	**Covered**	Not Executed	Prevents DDE execution and CSV structure corruption
S-SR-003	Over-Length Buffer Safety: accept input of any length safely, rejecting > max bounds	TC-SEC-03	TC-SEC-03	TC-SEC-03_OverLengthInputSafety	tests/test_over_length_input_safety.cpp	Sprint 2	Issue #49 / PR #111	Security	System	10,000+ character input rejected; application continues normally with zero corruption	**Covered**	Not Executed	Guarantees hea safety against buffer overruns
S-SR-004	Untrusted CSV Data Validation: validate loaded fields and skip invalid records	TC-SEC-04	TC-SEC-04	TC-PER-03_SkipMalformedCSV	tests/test_skip_malformed_csv.cpp	Sprint 1 / 2	Issue #50 / PR #79, #112	Security / Robustness	Integration	Malformed file rows skipped; only valid records load into system memory	**Covered**	Not Executed	Discrepancy: Exercised unde TC-PER-03 and TC-NFR-02
S-SR-005	Non-Destructive File Writes: preserve previous file content if disk write fails	TC-PER-07	TC-PER-07	TC-PER-07_HandleCSVWriteFailures	tests/test_csv_write_failures.cpp	Sprint 1 / 2	Issue #51 / PR #89, #113	Security / Reliability	Integration	Simulated write failure preserves previous file completely intact and byte-identical	**Covered**	Not Executed	Official STP Tab 7 test ID is TC-PER-07 (ali TC-SEC-05 in sprint notes). Atomic non-destructive write logic verifi via CTest target C-PER-07_Han CSVWriteFailur (PR #113).

q ID	Requirement Description	STP Test Case ID	Updated Test Case ID	Current CTest Target	Test Source File	Sprint	Issue / PR	Test Type	Test Level	Expected Result	Coverage Design Status	Execution Status	Remarks / Mapping Notes
S-SR-006	Non-Deletion Audit Integrity: retain cancelled appointments in CSV with Cancelled status	`TC-APT-09`	`TC-APT-09`	`TC-SEC-06_RetainCancelledAppointmentRecords`	`tests/test_retain_cancelled_appointment_records.cpp`	Sprint 2	Issue #52 / PR #114	Security / Audit	System	Cancelled appointment retained in appointments.csv with status Cancelled; record count preserved	**Covered**	Not Executed	Official STP Tab 7 test ID is TC-APT-09 (alia TC-SEC-06 in sprint notes). Dedicated secu CTest target TC EC-06_RetainC celledAppointm Records added PR #114.
S-SR-007	Patient Data Privacy Minimization: conceal address, contact, and notes in appointment views	`TC-SEC-07`	`TC-SEC-07`	`TC-SEC-07_MinimizePatientDataInAppointmentViews`	`tests/test_minimize_patient_data_in_appointment_views.cpp`	Sprint 2	Issue #53 / PR #115	Security / Privacy	System	Appointment views exclude contact number, address, and medical notes; show only ID and Name	**Covered**	Not Executed	Enforces least-privilege information disclosure
S-SR-008	Air-Gapped Offline Operation: zero network socket activity or external data transmission	`TC-NFR-09`	`TC-NFR-09`	`TC-NFR-09_StandaloneOffline`	`tests/test_standalone_offline.cmake`	Sprint 2	Issue #54 / PR #105, #116	Security / Isolation	System	Binary symbols and runtime execution verify zero network sockets or external telemetry	**Covered**	Not Executed	Official STP Tab 7 test ID is TC-NFR-09 (alia TC-SEC-08 in sprint notes). Verified via objdump networ symbol inspecti in test_standalo _offline.cmake.

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3. Part B — Additional Sprint Scope Requirements Traceability Matrix (Supplementary Scope)

The table below catalogs supplementary capabilities, non-functional enhancements, and security controls delivered during Sprint 1 and Sprint 2 beyond the original 54 baseline requirements.

Supplementary Scope ID	Requirement Description	Sprint	Issue #	PR #	Supplementary Test Case ID	Current CTest Target	Source Test File	Expected Result	Coverage Design Status	Remarks / Implementation Notes
HMS-SR-009	Filesystem Error Message Sanitization: omit raw filesystem paths, exception traces, or raw malformed rows from console errors	Sprint 2	#55	#117	`TC-SEC-09`	`TC-SEC-09_SanitizeErrorMessages`	`tests/test_sanitize_error_messages.cpp`	Error messages state reason without revealing OS paths, file paths, or exception text	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Prevents information disclosure.

Supplementary Scope ID	Requirement Description	Sprint	Issue #	PR #	Supplementary Test Case ID	Current CTest Target	Source Test File	Expected Result	Coverage Design Status	Remarks / Implementation Notes
HMS-SR-010	Restricted Data Directory Access: restrict all file operations strictly to designated data folder; reject arbitrary paths	Sprint 2	#56	#118	`TC-SEC-10`	`TC-SEC-10_RestrictDataDirectory`	`tests/test_restrict_data_directory.cpp`	System rejects directory traversal or external file paths; operates strictly in data directory	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Prevents path traversal.
HMS-F-022-EXT	Appointment Filtering Entity Existence Validation: verify patient/doctor exists before querying appointments when filtering	Sprint 2	#121	#123	`TC-APT-12`	`TC-APT-12_FilterUnknownEntity`	`tests/test_filter_unknown_entity.cpp`	Displays "Patient not found" or "Doctor not found" on unknown ID; "No appointments found." only for registered entity with zero appointments	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Extension of HMS-F-022 and UC-08 exception 3a; 69 assertions.
HMS-F-001/008-EXT	Duplicate Patient and Doctor Identity Validation: detect duplicate human individuals via normalized name, contact, specialization	Sprint 2	#124	#125	`TC-ID-01`	`TC-ID-01_DuplicateIdentityValidation`	`tests/test_duplicate_identity_validation.cpp`	Detects duplicate identity heuristics; displays existing ID; requires explicit user confirmation (Y/N) before saving	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Supplementary data integrity guard for patient and doctor registration.
HMS-F-023-EXT	Appointment Cancellation Confirmation Workflow: display appointment summary before confirmation; validate Y/N response	Sprint 2	#120	#122	`TC-APT-11`	`TC-APT-09_CancelAppointment, TC-APT-10_RejectInvalidCancellation`	`tests/test_cancel_appointment.cpp, tests/test_reject_invalid_cancellation.cpp`	Displays appointment details (Patient, Doctor, Date, Time, Status) and confirms Y/N before cancelling; retry on invalid input	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Supplementary functional enhancement to UC-09 cancellation workflow.
HMS-F-003/010/018-EXT	Universal Immediate Field Validation and Interactive Retry Handling: validate each field immediately and re-prompt same field on error	Sprint 2 / Post-Sprint	#119	#127	`TC-VAL-01`	`No dedicated CTest target; exercised in TC-APT-01, TC-APT-09, TC-DOC-01, TC-PAT-05, etc.`	`tests/test_book_appointment.cpp, tests/test_cancel_appointment.cpp, tests/test_add_doctor.cpp, tests/test_patient_details_and_list.cpp`	Field-level validation error displays immediately; prompts for retry of same field without losing previously entered fields	**Covered**	Supplementary Project Scope ID — Not part of the original approved SRS/STP baseline. Interactive console UX enhancement implemented via readValidatedField template.

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4. Traceability Metrics & Analysis Summary

A. Baseline Scope Coverage Analysis (Denominator = 54)

Requirement Category	Total Baseline Requirements	Fully Designed & Mapped	Partially Mapped	Unmapped	Design Coverage %
Functional Requirements (HMS-F)	36	36	0	0	**100.0%**
Non-Functional Requirements (HMS-NF)	10	9	1 (`HMS-NF-007`)	0	**100.0%**
Security Requirements (HMS-SR)	8	8	0	0	**100.0%**
Total Baseline Requirements	**54**	**53**	**1**	**0**	**100.0%**

Note on Partial Mapping: `HMS-NF-007` (Cross-Platform Compilation) is fully verified at build time through CMake compile flags (`-std=c++17 -Wall`) and multi-compiler support, but has no dedicated executable target in CTest. It is documented honestly as partially mapped to CTest targets.

B. Supplementary Sprint Scope Summary

- **Additional Security Requirements:** 2 (`HMS-SR-009`, `HMS-SR-010`).

- **Additional Functional Extensions:** 4 (`HMS-F-022-EXT`, `HMS-F-001/008-EXT`, `HMS-F-023-EXT`, `HMS-F-003/010/018-EXT`).

- **Total Supplementary Scope Items: 6**.

- **Supplementary Design Coverage: 100.0%** (all 6 mapped to merged PRs and test suites).

C. Test Target Summary

- **Total Automated CTest Targets in Suite: 51**.

- **CTest Targets Mapping to Baseline Scope:** 47.

- **CTest Targets Mapping Directly to Supplementary Scope:** 4 (`TC-SEC-09_SanitizeErrorMessages`, `TC-SEC-10_RestrictDataDirectory`, `TC-APT-12_FilterUnknownEntity`, `TC-ID-01_DuplicateIdentityValidation`). (Note: Supplementary items `HMS-F-023-EXT` and `HMS-F-003/010/018-EXT` are exercised within existing interactive targets such as `TC-APT-09`, `TC-APT-10`, `TC-PAT-03`, `TC-DOC-03`, and `TC-APT-01`).